

Tech Stack (2026-2027)

Addendum for LoppisFynd - Comprehensive Review

Client (Flutter)

- Flutter app for iOS and Android
- State management + DI: Riverpod
- Offline-first persistence + caching: Drift (SQLite)
- Barcode scanning: Google ML Kit
- Background jobs: workmanager
- Observability: Sentry (crash + performance)

AI / ML

- Default identification: Cloud Gemini (vision) via server-side proxy (no keys in app)
- Offline fallback: YOLOX (Apache-2.0; BaseDetection/YOLOX) using a small model (<10MB target)
- Unified response contract: AiResult JSON used by all backends and stored in Drift
- Privacy: crop + compress before upload; "send crops only" toggle enforced

Backend (Supabase)

- Auth: Supabase Auth
- Database: Postgres (sync + user data)
- Edge Functions:
 - - /ai/identify proxy (Gemini calls, quotas, rate limits, kill switch)
 - - existing tradera-proxy pattern for market comps
- Storage (optional): Supabase Storage for user exports / shared assets

Quality / Delivery

- CI: format + analyze + custom_lint + tests + release builds
- Feature flags: kill switch for cloud identify, staged rollout for risk control
- Integration tests focused on first run + capture/identify + settings toggles

Note: Moving away from Gemma means no multi-GB model downloads on first run.